BEAUTRIUM ORDER MANANAGEMENT SYSTEM

Business & System Requirement Specification

Product Scope: Order Consolidation + Order Picking Productivity

Product concept
One Beautrium Order Mananagement System with a Shared Order Core and two Product Scope modules. Order Consolidation optimizes the picking approach; Order Picking Productivity controls assignment, execution, productivity, SLA and backlog.

Document Information	Details
Document Type	Business + System Requirement Specification
Version	1.1 - Product Requirement Revision
Prepared For	Warehouse Operation / Development Team
Product Scope	1) Order Consolidation 2) Order Picking Productivity
Order Source	WMS order file / future API
Order Grain	Order Header + Order Line (SKU + Bin Code + QTY)
Zone Source	Bin Code joined to Location Master
Productivity Grain	Order level
Data Retention	High-volume transactional data: rolling 7 days (Week -1) after successful weekly productivity export. Master/config/user data retained.
Design Capacity	Minimum 5,000 orders/day with growth buffer
Status	Development-ready product requirement; delivery sequencing is determined by the implementation team

Key design principle

- WMS is the source of truth for the actual pick Bin Code. The application does not decide Shelf vs Rack allocation.
- The same imported WMS Order Header and Order Line dataset is the single source of truth for both Product Scope modules; duplicate application databases are not permitted.
- Operational thresholds, matching limits, retention days and scheduled integrations must be configuration-driven and must not require code changes.
- An Order can touch multiple Zones because different lines may have different Bin Codes; productivity remains tracked at Order level.

0. Document Control & Contents

Version	Change Summary	Owner / Approver
1.1	Reframed as Beautrium Order Mananagement System with two Product Scope modules; added Front-end/CSS standard, weekly Google Sheets productivity export, and Week -1 transaction retention.	Warehouse Operation / Product Owner

1. Executive Summary

Beautrium Order Mananagement System เป็น Product เดียวที่ใช้ Order dataset จาก WMS ร่วมกัน โดยมี 2 Product Scope ที่ทำหน้าที่ต่างกันแต่เชื่อมโยงกันบน Shared Order Core เดียว ได้แก่ Order Consolidation และ Order Picking Productivity เพื่อลดข้อมูลซ้ำ ลด Logic ซ้ำ และให้สถานะ Order เป็นแหล่งข้อมูลเดียวกันทั้งระบบ

Product Scope	Business Question	Primary Functions
Order Consolidation	How should we pick to reduce walking?	Pre-screen, SKU matching, batch split, pick sequence, consolidated pick report, pick & sort
Order Picking Productivity	Who is doing the picking work, how fast, and is operation on track?	Assignment, workload control, order completion, KPI, backlog, Zone dashboard, Control Tower

Critical data rule Every WMS order line already contains the real Bin Code to pick. If the same SKU must be picked from Shelf and Rack, WMS sends two lines; the system preserves both lines and joins each Bin Code to Location Master for Zone and Pick Sequence.

The shared platform avoids duplicate import, duplicate user management, inconsistent order status and conflicting master data. Consolidation batches and productivity assignment batches remain separate entities because they have different business purposes, but they can be linked.

2. Objectives & Success Criteria

- Reduce repeated walking by grouping orders with similar SKU demand and producing one physical walking sequence.
- Use yesterday as the default Order Date for Order Consolidation analysis while retaining historical dates for audit and reports.
- Allow supervisors to understand daily matching potential, unmatched reasons and batch quality before releasing pick reports.
- Track picker productivity at Order level while showing workload contribution by Zone based on actual WMS Bin Code.
- Control assignment workload around configurable target pieces per hour/batch (initial benchmark around 300 pieces).
- Provide Warning / Overdue / Critical monitoring, backlog handling, Short Pick, Admin verification and historical analysis.
- Keep masters, thresholds, permissions and operational rules configurable without code changes.
- Provide PC-first supervisor/admin UX and PDA/mobile-friendly execution screens for Order Picking Productivity.

3. Product Scope

Component	Scope	Key Deliverable
Shared Order Core	Authentication, users, WMS order import, order header/line, location master, configuration, audit	One source of truth for both Product Scope modules
Order Consolidation	Pre-screen, P1-P4 matching, batch split/review, A4 consolidated pick report, pick & sort support	Matching analysis + operational consolidation pick report
Order Picking Productivity	Assignment, workload control, picker completion, admin verification, backlog, Zone Dashboard and Control Tower	Order-level productivity + operational control dashboards
Integration Services	Google Sheets weekly export, scheduled housekeeping, future WMS API/PDA integrations	Reliable data movement and retention control

In Scope

- Excel/CSV import and future WMS API
- Order-line storage including Bin Code
- Duplicate/upsert control
- Consolidation matching and report generation
- Location Master with Zone and Pick Sequence
- Assignment and productivity tracking
- Backlog / alerts / dashboards / exports
- User / role / audit / configuration

Out of Scope for current combined release

- Stock movement or stock adjustment in WMS
- Replenishment calculation/transaction
- Changing WMS allocation or choosing Shelf vs Rack
- Replacing core WMS
- Automated labor forecasting in initial delivery stages

4. Core Architecture & Shared Components

Recommended logical architecture:

> WMS → Shared Order Core → Two Product Scope Modules WMS Import/API feeds a single Order Header / Order Line store. Order Consolidation reads the data for consolidation. Order Picking Productivity reads the same order entities for assignment, completion and productivity. Location Master enriches every order line with Zone and Pick Sequence.

Shared Component	Purpose
Authentication / User / Role	Single identity and permission model across all modules
Order Import Service	Validate, deduplicate, upsert and log WMS files
Order Core	Order Header + Order Line source for both delivery stages
Location Master	Bin → Zone / Aisle / Side / Bay / Level / Block / Pick Sequence
Configuration Service	Effective-dated parameters; no hard-coded thresholds
Audit / Status History	Trace all critical changes and operational decisions
Reporting Service	Excel export, A4 PDF/print rendering, historical query

5. WMS Input / Order Core Data Model

The order file is line-level. Bin Code is mandatory because it represents the actual WMS-directed pick location.

Field	Level	Requirement / Rule
Order No	Header	Required; unique within warehouse/source key
Original Order Date	Header	Required; Order Consolidation only compares orders from the same date
Store Code	Header	Required; distinct store count used in consolidation
SKU	Line	Required
Bin Code	Line	Required; actual pick location from WMS
QTY	Line	Required; planned pieces for that SKU/Bin
Priority / Cut-off	Header	Optional but recommended
Cancel Status	Header	Optional but recommended
Item Description	Line	Optional; useful on print report
Barcode values	Line/Master	SKU barcode and Bin barcode if supplied; otherwise generated from code for print representation

Example - same SKU from two physical locations:

Order	SKU	Bin Code	QTY
ORD001	A001	B015	5
ORD001	A001	R020	10

Data interpretation ORD001 is one order and A001 is one unique SKU for matching, but there are two physical pick lines. Consolidated pick output must group by SKU + Bin Code, never by SKU alone.

Duplicate / Update Control

- Recommended unique import key: Warehouse + Order No + Original Order Date, or a guaranteed WMS key.
- Order line unique key should include Order + SKU + Bin Code + source line identifier where available.
- Re-upload must not duplicate QTY. Allowed updates should create change history.
- Invalid Bin Code must go to Error Queue and cannot be released to a pick report until Location Master is resolved.

6. Location Master, Zone & Pick Sequence

Division is removed from the combined design. Zone is a property of physical Bin Code through Location Master.

Location Master Field	Rule
Bin Code	Primary business key from WMS
Zone Code / Zone Name	Report and workload grouping
Aisle	Physical walking hierarchy
Side	A-Z side identifier
Side Pair	AB, CD, EF ...

Location Master Field	Rule
Side Direction	Odd letter = RIGHT; even letter = LEFT
Bay	Natural numeric order
Level	Alphabetic/numeric order
Block	Natural numeric order
Pick Sequence	Composite numeric key used to sort physical walking
Active Flag	Inactive location cannot be used for released report

Physical walking sort order

Aisle → Side Pair → Bay → Right → Left → Level → Block Example A1-B-01-A-01 = 01-01-01-2-01-01, where side direction 1 = RIGHT and 2 = LEFT. The composite key is business-readable and easier to maintain than a pure running number.

7. User Roles & Permissions

Role	Order Consolidation	Order Picking Productivity	Administration
System Admin	Full	Full	Users, masters, config, audit
Warehouse Manager / Control Tower	View / approve	All dashboards / monitoring	Read config/audit
Supervisor	Generate/review/release batch	Assign / monitor / override / verify	Operational config where authorized
Planner / Admin	Import / matching / report	Create assignment / verify	Limited operational master
Zone Controller	View matching impact by zone	Zone dashboard / drilldown	No master edit
Picker	View/print task if needed	Own work / close order / short pick	None
Viewer	Read only	Read only	None

Permission principle Menu visibility and action permission are separate. A user may view a module but not approve, override, release or edit configuration.

8. End-to-End Business Workflow

1) Import WMS order lines and validate / deduplicate.

2) Enrich each line from Location Master: Zone + physical Pick Sequence.

3) Order Consolidation uses Order Date (default yesterday) to pre-screen eligible orders and generate matching candidates.

4) Supervisor reviews P1-P4 candidates, batch splits and exceptions; approved batches produce A4 Consolidated Pick Reports.

5) Picker performs Pick & Sort by report sequence. At Consolidation Area, WMS Picking-by-Order is used to confirm each order from the already-picked pool.

6) Order Picking Productivity assigns orders/work packages to a picker and controls planned pieces against configurable workload target.

7) Picker completes each Order; KPI stops at Picker Completed Time. Short Pick captures actual/short/reason.

8) Admin verifies and Final Closes or Rejects for correction.

9) Dashboards show Order productivity, Zone workload contribution, backlog, alerts and Control Tower status.

9. Order Consolidation

Purpose: reduce walking by combining orders with similar SKU demand, while preserving the existing WMS Order → SKU → QTY execution model.

Operating model

- System generates a Consolidated Pick Report only; no PDA allocation is required.
- Picker picks total quantity by physical location according to Pick Sequence.
- Picked goods move to Consolidation Area.
- User opens existing WMS Picking-by-Order one order at a time and takes items from the consolidated pool instead of walking back to real locations.
- WMS remains the final source of order confirmation/closure.

Future automation may assign generated consolidation batches directly to PDA; this is an optional product enhancement and is not required for the report-based operating model defined here.

10. Order Consolidation - Pre-screen, Matching & Batch Split Rules

The backend uses Adaptive Pre-screen + Fixed Rule Matching. It may optimize candidate search strategy based on the daily order population, but business thresholds do not change automatically.

10.1 Eligibility Pre-screen

Rule	Decision
Order Date	Compare only orders with the same Original Order Date
Order > Maximum Unique SKU	Exclude from consolidation; route to Single Order
Cancelled / invalid / missing Bin	Exclude / error queue
Unique SKU Count	Count SKU once per order even when same SKU appears in multiple bins
Exact SKU Signature	Same SKU set → P1 candidate directly
Non-exact candidates	Use SKU inverted index to avoid all-to-all comparisons

10.2 Matching Priority

Priority	Match Rule	Minimum Pieces	Action
P1	100% Perfect SKU match	Config (initial >50)	Create candidate
P2	≥ Config P2 threshold (initial 80%)	Config (initial >50)	Create candidate
P3	≥ Config P3 threshold (initial 50%)	Config (initial ~80)	Create candidate
P4	≥ Config P4 threshold (initial 30%)	Config (initial >150)	Create candidate
P5	Does not qualify	-	Single Order

Recommended symmetric pair match = Common Unique SKU / MAX(Unique SKU count of Order A, Order B). This prevents a 2-SKU order from being treated as a perfect match to a 10-SKU order.

10.3 Configurable Batch Capacity

Parameter	Purpose
Minimum Stores	Minimum distinct stores required to release a batch
Target Stores	Preferred split size
Maximum Stores	Hard maximum distinct stores per batch

Parameter	Purpose
Maximum Orders	Optional independent cap because one store may contain multiple orders
Maximum Unique SKU	Eligibility/cap; no mandatory minimum Unique SKU
Minimum Pieces by Priority	Batch economics / workload threshold

10.4 Batch Splitting

Example If 20 stores are Perfect Match and Target Stores = 7 / Maximum Stores = 8, split to 7 + 7 + 6. Every resulting batch must independently pass the Minimum Pieces rule; redistribute stores to balance pieces where possible.

- Orders selected into a released candidate must be locked against duplicate use.
- Batch split should balance Store Count + Pieces, not Store Count alone.
- If the final remainder cannot meet minimum criteria, keep it waiting for later matching or route to Single Order according to operational cut-off.

11. Order Consolidation - Consolidation Pick Report

The report is designed for A4 printing, compact layout, low visual clutter, and direct physical execution.

Required Element	Design Requirement
Header	Warehouse, Order Date, Batch No, Priority, generated time, supervisor, batch barcode
Summary	Stores, Orders, Unique SKU, Total Pieces, Zones covered
Pick Lines	Sorted by Location Master Pick Sequence
Grouping key	SKU + Bin Code; sum QTY across orders in batch
Bin Barcode	Mandatory printable barcode + human-readable Bin Code
SKU Barcode	Mandatory printable barcode + human-readable SKU
Columns	Seq, Zone, Aisle/Side, Bin, SKU, Description, Total Pick Qty, Stores, Orders
Footer	Pick instructions, consolidation instructions, page x/y

Important A SKU appearing in B015 and R020 must produce two report rows even if it is the same SKU, because WMS has assigned two real pick locations.

Mockup - A4 Consolidation Pick Report with Bin and SKU barcodes

ORDER CONSOLIDATION

Consolidation Pick Report

Sprint 1 – Matching & Pick

Warehouse	Bangna DC	Batch No.	BATCH-250520-001
Report Date	20 May 2025	Generated Time	20 May 2025 09:30
Order Date	Yesterday	Supervisor	Somsak P.

BATCH-250520-001

Total Stores	Total Orders	Total Unique SKU	Total Pieces	Zones Covered
7 Stores	260 Orders	126 SKUs	28,451 Pieces	6 Zones

Route / Zone Summary

Zone A (512 Locations) → Zone B (336 Locations) → Zone C (198 Locations) → Zone D (142 Locations) → Zone E (60 Locations) → Zone F (0 Locations)

Seq	Zone	Aisle	Side	Bin Code	Bin Barcode	SKU	SKU Barcode	Item Description	Pick Qty	Stores	Orders
1	A	A-01	L	A-01-L-01	A01L01	SKU-000123	000123	Screw M6 x 20	360	5	18
2	A	A-01	R	A-01-R-01	A01R01	SKU-000456	000456	Washer M6	520	6	24
3	A	A-02	L	A-02-L-01	A02L01	SKU-000789	000789	Nut M6	420	4	16
4	B	B-01	L	B-01-L-01	B01L01	SKU-001234	001234	Bolt M8 x 30	300	5	17
5	B	B-01	R	B-01-R-01	B01R01	SKU-001567	001567	Spring Washer M8	260	4	12
6	B	B-02	L	B-02-L-01	B02L01	SKU-001890	001890	Hex Nut M8	400	6	19
7	C	C-01	L	C-01-L-01	C01L01	SKU-002345	002345	Flat Head Screw M4	180	3	9
8	C	C-01	R	C-01-R-01	C01R01	SKU-002678	002678	Plastic Anchor M6	210	4	11
9	D	D-01	L	D-01-L-01	D01L01	SKU-003456	003456	Cable Tie 3.6 x 200	600	6	22
10	D	D-01	R	D-01-R-01	D01R01	SKU-003789	003789	Electrical Tape 20m	320	5	15
11	E	E-01	L	E-01-L-01	E01L01	SKU-004567	004567	Label 50x30mm	240	4	12
12	E	E-01	R	E-01-R-01	E01R01	SKU-004890	004890	Marker Pen Black	160	3	8

Pick Instructions

- Follow the sequence (Seq) and route by Zone.
- Scan Bin Barcode and SKU Barcode when picking.
- Pick the exact quantity shown in Total Pick Qty.
- Place picked items in pick tote with zone label.

Sort at Consolidation Area

- Sort by **Store** as shown on the tote label.
- Verify quantity and SKU before packing.
- Attach packing list and hand over to Packing Team.
- Report any discrepancy to the Supervisor.

Notes

Priority Legend: P1 High | P2 Medium | P3 Low

12. Order Picking Productivity - Assignment & Execution

Productivity is tracked at Order level. An Order may touch one or more Zones through its lines; Zone is not an Order ownership field.

12.1 Assignment Batch

- Admin scans/selects Orders to create an Assignment Batch.
- System sums planned pieces in real time.
- Workload thresholds are configurable (initial benchmark around 300 pieces / one-hour work package).
- Timer starts only when Admin confirms assignment. Picker does not scan to accept work.
- Assignment batch may include Orders that touch multiple Zones; the previous cross-zone restriction is removed.
- A Consolidation Batch may be linked as the source work package, and several consolidation/single orders may be included subject to workload rules.

Workload Level	Illustrative Default	Display
Low	<270 pieces	Yellow
Target	270-300	Green
Acceptable Over	301-330	Yellow
Over Capacity	>330	Red / warning

All numeric values above must be configuration-driven.

12.2 Picker Completion

- Picker closes each Order by scan/button.
- Complete 100% or Short Pick.
- Short Pick captures Actual Picked Pieces, Short Pieces, Reason and Remark.
- Picker Completed Time stops the Order-level picking KPI.
- Admin waiting time is not included in Picker productivity.

12.3 Consolidation interaction

When orders are physically picked as a consolidation batch, all constituent orders can inherit the assignment start time. Each order records its own completion time when it is confirmed during consolidation / WMS order-by-order processing. Batch-level KPI is retained in parallel for fairness and operational analysis.

13. Order Picking Productivity - Status, Backlog & KPI Rules

Status	Definition
New	Imported and available
Assigned	Assignment confirmed; timer started
In Progress	Picker working
Warning	Elapsed threshold reached (config)
Overdue	Overdue threshold reached
Critical	Critical threshold reached
Picker Completed 100%	Picker closed complete
Picker Completed Short	Picker closed with short pick
Waiting Admin Verification	Picker completed; admin pending
Admin Rejected	Returned for correction
Correction in Progress	Rework
Final Closed 100%	Admin confirmed complete
Final Closed Short	Admin confirmed short
Picking Backlog	Not Picker Completed by operational end-of-day
Verification Backlog	Picker completed but not admin confirmed
Cancelled	Cancelled by source/system rule

KPI formulas

KPI	Formula / Rule
Order Cycle Time	Picker Completed Time - Assigned Time
Actual Pieces / Hour	Actual Picked Pieces / productive cycle hours
Completed Orders / Hour	Picker Completed Orders / productive hours
Picking SLA	Orders completed within configured SLA / completed orders
100% Completion Rate	100% completed orders / picker-completed orders
Short Pick Rate	Short Pick orders / picker-completed orders
Batch SLA	Assignment/Consolidation batch completed within configured SLA
Idle Time	Next Assigned Time - previous batch completed time

Backlog Rules

- Backlog stores Original Order Date, Backlog Date, Completion Date and Backlog Age.
- Productivity is credited to the date Picker Completed, even when the order began earlier.
- Control Tower Total Backlog = Picking Backlog + Verification Backlog, with both types visible separately.

14. Zone Dashboard & Control Tower

Zone dashboard is location-driven while productivity remains Order-driven. An Order can appear in more than one Zone contribution if its lines are physically located in multiple Zones.

14.1 Zone Dashboard

KPI / View	Definition
Orders Touching Zone	Distinct Orders with at least one line/Bin in the Zone; non-additive across Zones
Planned Pieces	Sum QTY for lines in the Zone
Completed / Remaining Pieces	Based on completion data available at line/order allocation level
Unassigned / Assigned / In Progress	Order states filtered to Orders touching the Zone
Warning / Overdue / Critical	Order alerts where the order touches the Zone
Picker Completed	Orders completed by picker
Active Pickers	Pickers whose current assignment contains work touching the Zone
Picker Monitor	Picker, current batch/orders, zones involved, times, pieces, productivity, status

Dashboard interpretation Orders Touching Zone must not be summed across Zone cards to obtain Total Orders. The Control Tower Total Orders is the unique Order count from Order Core.

14.2 Control Tower

- Total Orders / Total Pieces / Completed / Remaining
- Total Backlog split Picking vs Verification
- Unassigned / Assigned / In Progress / Warning / Overdue / Critical / Picker Completed
- Waiting Admin / Final Closed / Rejected
- Overall SLA / Avg Cycle Time / Pieces per Hour / Active Pickers / Short Pick
- Zone status cards with Green / Yellow / Red thresholds and drilldown
- Daily/weekly trend and top delayed orders/pickers

15. Front-end UX/UI & CSS Design Standard

Front-end navigation must follow the operational sequence so users move naturally from Order Import → Consolidation Analysis → Pick Report → Work Assignment → Productivity Monitoring → Reporting / Administration.

Menu Group	Menu	Primary Users
Overview	1. Operations Dashboard	Manager / Supervisor
Order Core	2. Order Pool / Import Status	Admin / Planner
Consolidation	3. Matching Dashboard	Supervisor / Planner
Consolidation	4. Matching Analysis & Batch Review	Supervisor / Planner
Consolidation	5. Consolidation Pick Report	Supervisor / Picker
Consolidation	6. Consolidation History	Supervisor / Viewer
Picking Control	7. Work Assignment	Admin / Supervisor
Picking Control	8. Picker Monitor	Admin / Zone Controller
Picking Control	9. Zone Dashboard	Zone Controller
Picking Control	10. Control Tower	Manager / Control Tower
Picking Control	11. Backlog Monitor	Admin / Manager
Analytics	12. Productivity / SLA / Short Pick	Manager / Supervisor
Administration	13. User Management	System Admin
Administration	14. Location Master	System Admin
Administration	15. Configuration / Audit	System Admin

Global UX rules

- Order Date defaults to Yesterday (D-1) on Order Consolidation screens. Database screens retain Week -1 transaction data; older productivity history is accessed from the weekly Google Sheets archive.
- Use consistent colors: Green = ready/normal, Yellow/Orange = attention, Red = overdue/critical/error, Purple = matching/analysis, Blue = navigation/action.
- KPI cards must be clickable for drilldown.
- Filters should persist during navigation within a module.
- Every destructive/override action requires confirmation and reason where applicable.
- Tables support search, sort, pagination, export and visible filter chips.
- Role-based menu hides irrelevant modules and actions.

15.1 Visual Direction

- Style: modern warehouse operations dashboard, clean/light content canvas, dark navigation, strong icon language, high information density without visual clutter.
- Beautrium identity: primary Red / Black / Gray / White. Semantic operational colors are allowed for status and matching so meaning remains immediately visible.
- Use one consistent icon library across the Product (Lucide, Material Symbols or equivalent SVG icon set). Do not mix unrelated icon styles.

15.2 CSS Design Tokens

Recommended CSS variables / design tokens:

--btm-red: #D71920; --btm-red-dark: #A50F15; --nav-bg: #15181E; --page-bg: #F5F6F8; --surface: #FFFFFF; --text: #1F2937; --muted: #6B7280; --border: #E5E7EB; --info: #2563EB; --success: #16A34A; --warning: #F59E0B; --danger: #DC2626; --match-analysis: #7C3AED;

- Typography: Noto Sans Thai, Arial, sans-serif. Base 14px; page title 24-28px/700; section heading 18-20px/700; table 12-14px; numeric KPI 28-36px/700.
- Spacing: 8px base grid. Main page padding 24px desktop / 16px tablet; card gap 16px; card padding 16-20px.
- Cards: background #FFFFFF, 1px border #E5E7EB, radius 10-12px, subtle shadow only for hierarchy. Avoid heavy gradients.
- Buttons: primary action uses Beautrium red; secondary uses neutral white/gray; analysis actions may use info blue/purple; destructive actions use danger red only.
- Status colors: Green = Normal/Ready/Completed, Yellow = Warning, Orange = Overdue/attention, Red = Critical/Error, Purple = Matching/analysis, Gray = Closed/Inactive.

15.3 Layout / Responsive Rules

- Desktop-first operating size: 1366×768 and 1440×900. Left navigation 220-240px expanded and 64-72px collapsed.
- Top application bar: page title, Order Date, Warehouse filter, notifications and current user. Order Consolidation Order Date defaults to Yesterday.
- KPI cards use 4-7 cards per row depending on viewport; cards wrap responsively. Tables must keep key identifiers frozen/sticky where practical.
- Picker/PDA screens are responsive and touch-friendly with minimum 44×44px interactive targets, high-contrast status chips and minimal typing.
- A4 print/report CSS uses @page size A4 portrait; 10-12mm margins; hide navigation/buttons when printing; barcode area must not scale below scanner-readable size.

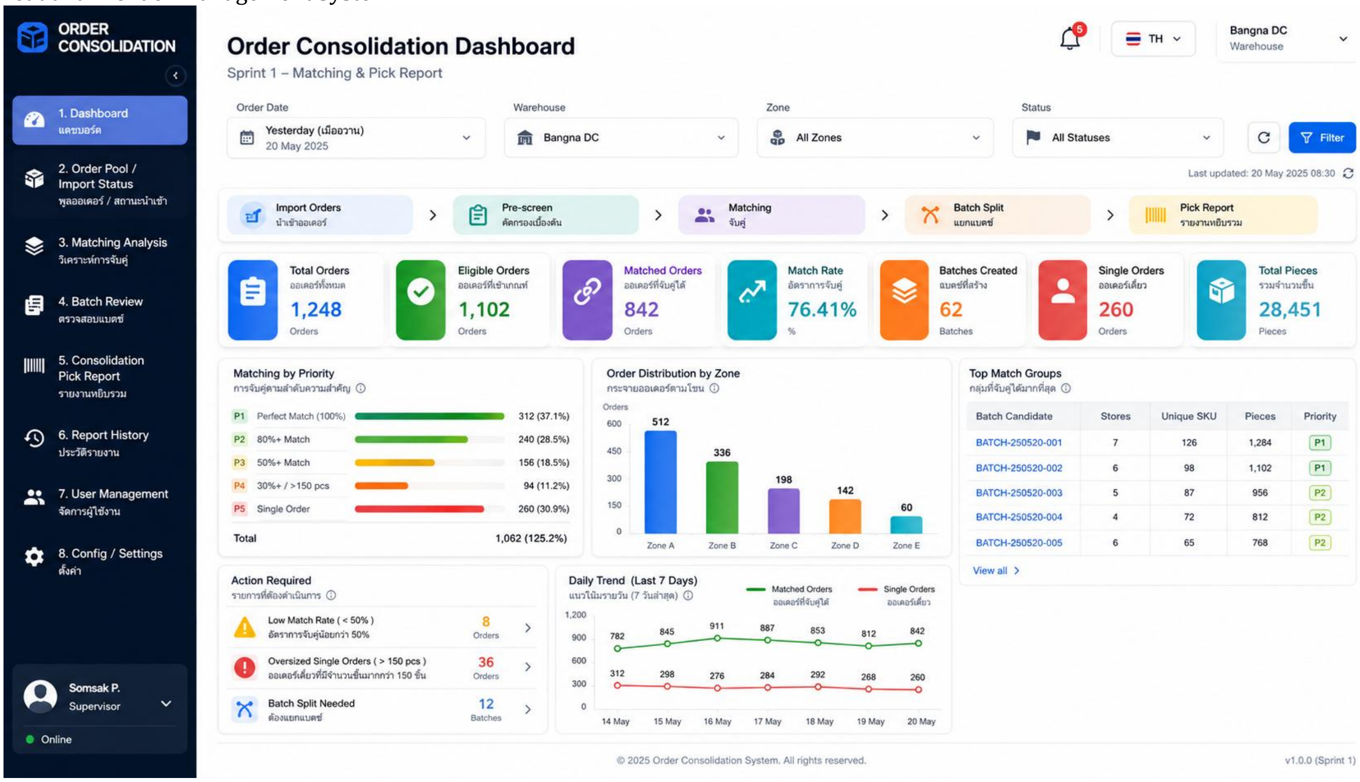
15.4 Reusable Front-end Components

- AppShell, SideNav, TopBar, PageHeader, FilterBar, KPICard, StatusChip, DataTable, EmptyState, AlertBanner, ConfirmDialog, AuditDrawer, BarcodeCell, ExportStatusCard, DatePresetPicker.
- All tables support server-side pagination for high volume, column sorting, search, visible filters, export where permitted and clear loading/error/empty states.
- Accessibility: keyboard focus visible; no status conveyed by color alone; text/icon label is required; contrast target WCAG AA for normal text.

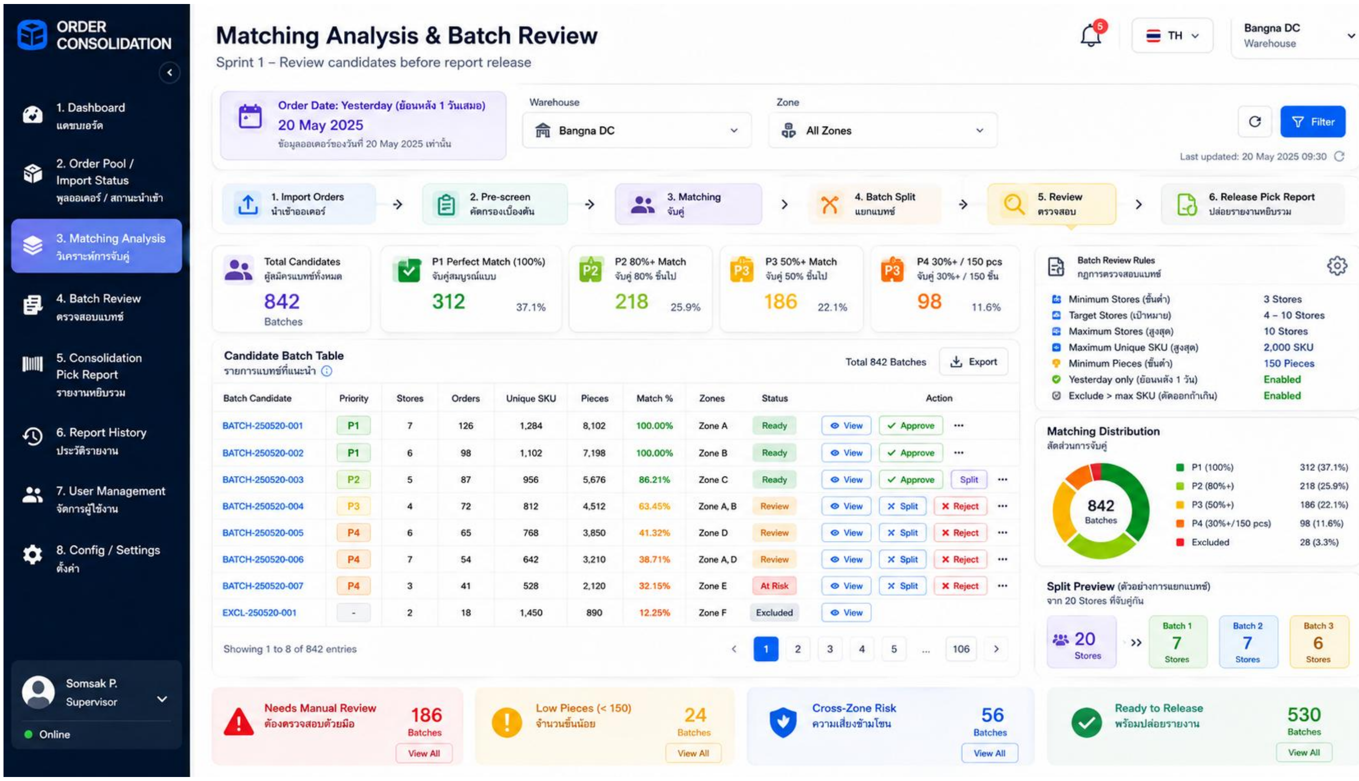
16. Mockup UX/UI

The following mockups are reference concepts; field labels and final counts must follow the production data model.

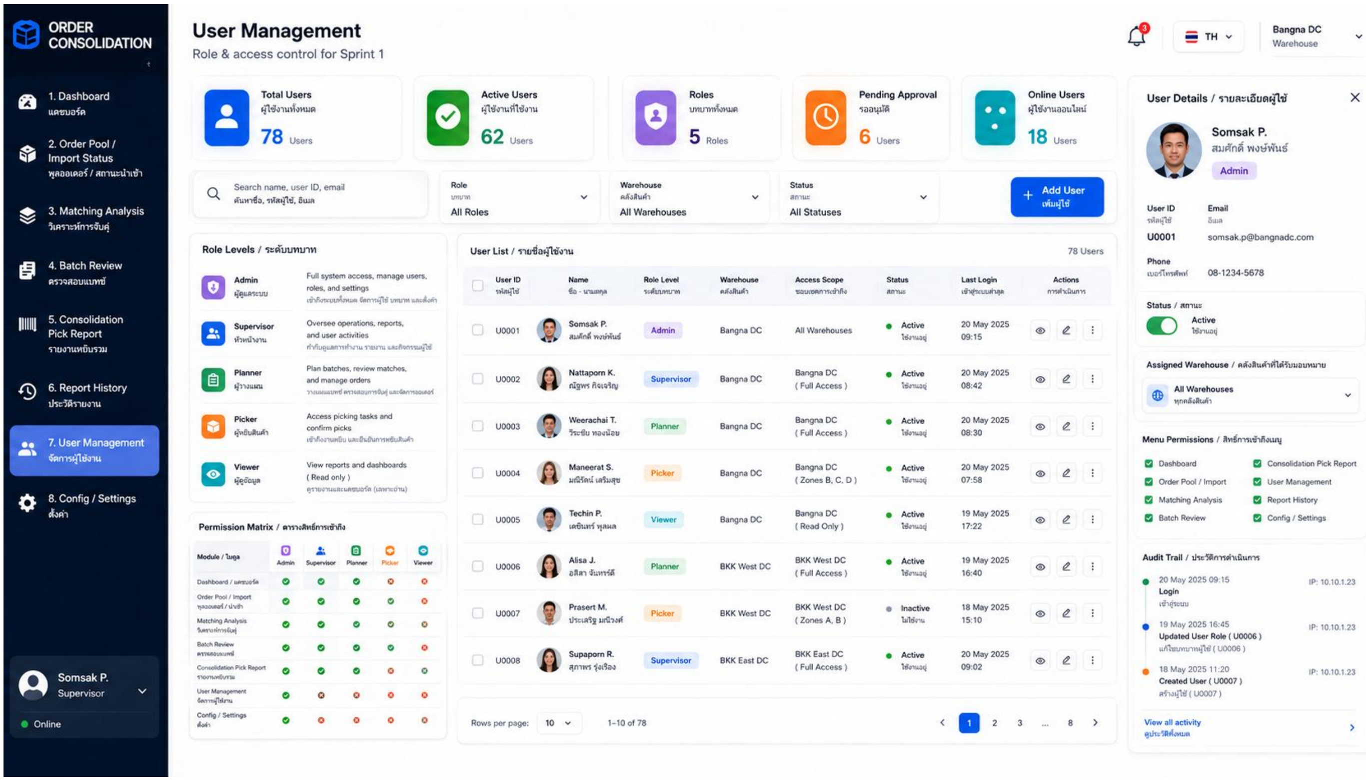
Mockup 1 - Order Consolidation Dashboard: daily matching potential, priorities, trend and actions


ORDER
CONSOLIDATION
1. Dashboard
แดชบอร์ด
2. Order Pool /
Import Status
พูลออเดอร์ / สถานะนำเข้า
3. Matching Analysis
วิเคราะห์การจับคู่
4. Batch Review
ตรวจสอบแบตช์
5. Consolidation
Pick Report
รายงานหยิบรวม
6. Report History
ประวัติรายงาน
7. User Management
จัดการผู้ใช้งาน
8. Config / Settings
ตั้งค่า
Somsak P.
Supervisor
Online
Order Consolidation Dashboard
Sprint 1 – Matching & Pick Report
5
TH
Bangna DC
Warehouse
Order Date
Yesterday (เมื่อวาน)
20 May 2025
Warehouse
Bangna DC
Zone
All Zones
Status
All Statuses
Filter
Last updated: 20 May 2025 08:30
Import Orders
นำเข้าออเดอร์
Pre-screen
คัดกรองเบื้องต้น
Matching
จับคู่
Batch Split
แยกแบตช์
Pick Report
รายงานหยิบรวม
Total Orders
ออเดอร์ทั้งหมด
1,248
Orders
Eligible Orders
ออเดอร์ที่เข้าเกณฑ์
1,102
Orders
Matched Orders
ออเดอร์ที่จับคู่ได้
842
Orders
Match Rate
อัตราการจับคู่
76.41%
%
Batches Created
แบตช์ที่สร้าง
62
Batches
Single Orders
ออเดอร์เดี่ยว
260
Orders
Total Pieces
รวมจำนวนชิ้น
28,451
Pieces
Matching by Priority
การจับคู่ตามลำดับความสำคัญ
P1 Perfect Match (100%) 312 (37.1%)
P2 80%+ Match 240 (28.5%)
P3 50%+ Match 156 (18.5%)
P4 30%+ / >150 pcs 94 (11.2%)
P5 Single Order 260 (30.9%)
Total 1,062 (125.2%)
Order Distribution by Zone
กระจายออเดอร์ตามโซน
Orders
600
450
300
150
0
512
336
198
142
60
Zone A
Zone B
Zone C
Zone D
Zone E
Top Match Groups
กลุ่มที่จับคู่ได้มากที่สุด
Batch Candidate | Stores | Unique SKU | Pieces | Priority
BATCH-250520-001 | 7 | 126 | 1,284 | P1
BATCH-250520-002 | 6 | 98 | 1,102 | P1
BATCH-250520-003 | 5 | 87 | 956 | P2
BATCH-250520-004 | 4 | 72 | 812 | P2
BATCH-250520-005 | 6 | 65 | 768 | P2
View all
Action Required
รายการที่ต้องดำเนินการ
Low Match Rate (< 50%)
อัตราการจับคู่น้อยกว่า 50%
8
Orders
Oversized Single Orders (> 150 pcs)
ออเดอร์เดี่ยวที่มีจำนวนชิ้นมากกว่า 150 ชิ้น
36
Orders
Batch Split Needed
ต้องแยกแบตช์
12
Batches
Daily Trend (Last 7 Days)
แนวโน้มรายวัน (7 วันล่าสุด)
Matched Orders
ออเดอร์ที่จับคู่ได้
Single Orders
ออเดอร์เดี่ยว
1,200
900
600
300
0
782 845 911 887 853 812 842
312 298 276 284 292 268 260
14 May 15 May 16 May 17 May 18 May 19 May 20 May
© 2025 Order Consolidation System. All rights reserved.
v1.0.0 (Sprint 1)

Mockup 2 - Matching Analysis & Batch Review: candidate approval, split preview, rules



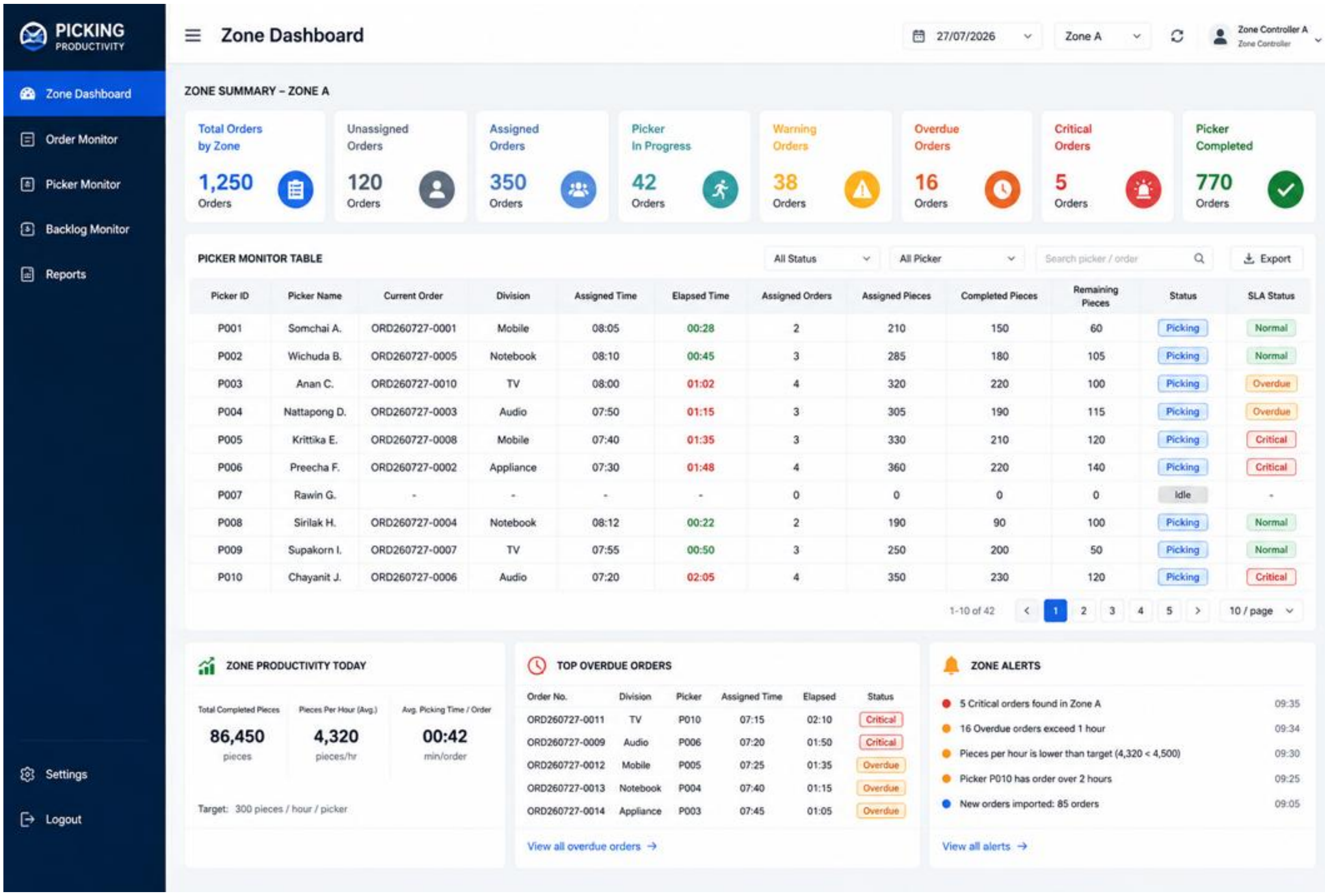
Mockup 3 - Shared User Management: role, scope, permissions and audit



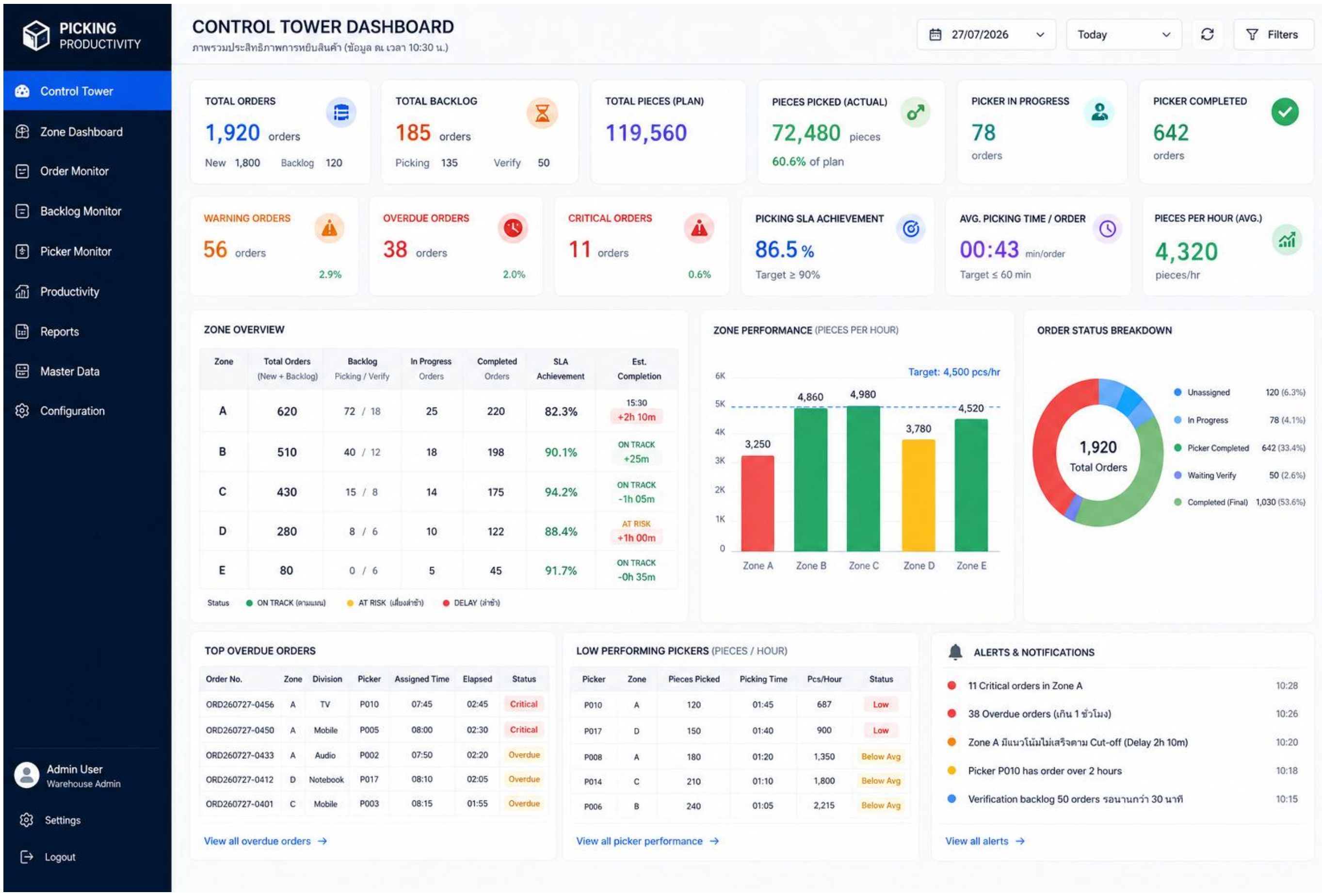
Beautrium Order Management System

Order Picking Productivity dashboard concepts from the existing requirement are retained as visual references. Zone logic uses WMS Bin Code joined to Location Master and never uses Division-based ownership.

Mockup 4 - Zone Dashboard reference (to be updated to Orders Touching Zone / location-based Zone)



Mockup 5 - Control Tower reference: total operation, backlog, Zone colors, alerts



17. Configuration & Master Data

Configuration	Scope / Example
Minimum Stores	Consolidation eligibility
Target Stores	Preferred batch split
Maximum Stores	Consolidation hard cap
Maximum Orders	Independent order cap
Maximum Unique SKU	Consolidation eligibility / cap
Minimum Pieces P1/P2/P3/P4	Priority-specific release threshold
Match % P2/P3/P4	Priority thresholds
Assignment target pieces	Initial benchmark 300
Assignment tolerance	Low/target/acceptable/over-cap ranges
Warning / Overdue / Critical minutes	Order SLA alert thresholds
Admin verification warning/overdue	Verification SLA
Zone color thresholds	Green/Yellow/Red
Operational day cutoff	Backlog calculation
Report barcode symbology	Code128/other implementation choice
Weekly productivity export enabled	Enable/disable scheduled Google Sheets export
Weekly export day/time/timezone	Schedule; default timezone Asia/Bangkok
Google Drive folder / naming pattern	Target folder and BTM_Productivity_YYYY-Www file naming
Transaction retention days	Default 7 days / Week -1; configurable
Purge schedule	Housekeeping execution schedule
Purge export safety gate	Enabled; do not purge covered week until export SUCCESS + reconciliation

Configuration governance

- Effective Date / Active Flag / Version
- Changed By / Changed At / Change Reason
- Old and new value audit
- Changes apply only to new batches/orders after effective activation; in-progress entities retain the configuration version used at creation.

18. Functional Requirements

ID	Requirement	Priority
FR-001	Single login and role-based menu/action permission	Must
FR-002	Import WMS line-level orders with Bin Code; show import summary	Must
FR-003	Deduplicate/upsert without duplicating QTY; change log allowed updates	Must
FR-004	Join every valid Bin Code to Location Master Zone and Pick Sequence	Must
FR-005	Default Order Consolidation Order Date to yesterday (D-1)	Must

ID	Requirement	Priority
FR-006	Pre-screen order eligibility using configurable Maximum Unique SKU	Must
FR-007	Generate P1 exact SKU signature candidates without pairwise calculation	Must
FR-008	Generate P2-P4 using configured symmetric SKU match thresholds	Must
FR-009	Split matching groups to store/order capacity and balance pieces	Must
FR-010	Lock orders to prevent duplicate released consolidation membership	Must
FR-011	Generate A4 Consolidated Pick Report sorted by Pick Sequence	Must
FR-012	Report prints Bin barcode + SKU barcode and separates same SKU across different bins	Must
FR-013	Admin creates Assignment Batch and sees pieces accumulated real-time	Must
FR-014	Configurable workload colors/thresholds and authorized override	Must
FR-015	Timer starts only when Admin confirms assignment	Must
FR-016	Picker closes each Order; complete or short pick	Must
FR-017	Admin verifies / confirms / rejects; waiting time excluded from picker KPI	Must
FR-018	Track productivity at Order level	Must
FR-019	Zone dashboard derives Zone from Order Line Bin Code, not Division	Must
FR-020	Control Tower uses unique Total Orders and separately shows Zone touching workload	Must
FR-021	Separate Picking and Verification Backlog	Must
FR-022	Weekly Productivity Export to Google Sheets with reconciliation, retry and export status	Must
FR-023	Scheduled purge keeps only rolling 7 days of high-volume transaction data; purge is blocked unless the required weekly export succeeded	Must
FR-024	Configuration changes do not require code deployment, including matching, workload, export schedule and retention days	Must
FR-025	Support linking consolidation batches to productivity assignment/work packages	Should
FR-026	Google Sheets export creates a new weekly spreadsheet in the configured Google Drive folder using order-level productivity data	Must
FR-027	System retains master data, user/role data and active configuration independently from the 7-day transaction purge	Must

19. Non-Functional Requirements

Category	Requirement
Performance	Common screens target <3 seconds under normal load; dashboard refresh target 15-60 seconds; batch generation should run asynchronously when population is large.
Capacity	Minimum design for 5,000 orders/day and line volume growth; no all-to-all matching across full population.
Retention	Rolling 7 days online data with archive/growth strategy.
Availability	Business-hours availability; backup and recovery supported.
Security	HTTPS, RBAC, password/SSO option, audit logging.
Data Integrity	Transactional batch lock/assignment/close operations; unique constraints.
Usability	PC-first planning/monitoring; responsive PDA/mobile execution for picker.
Browser	Current Chrome / Edge; responsive web.
Maintainability	Configuration-driven thresholds and master data.
Observability	Application logs, import error logs, job monitoring, DB metrics and alerting.
Print	A4 report must render consistently with barcode quiet zones and readable text.
Data Retention	High-volume transactional database retains only rolling 7 days by default; purge is scheduled and export-gated.
External Export	Weekly Google Sheets export must be idempotent/re-runnable and reconcile control totals before success.
Frontend Standard	Central CSS/design-token layer; responsive desktop/PDA layouts; reusable component library; print-specific A4 stylesheet.
Housekeeping	Scheduled jobs must be observable with last run, next run, status, affected rows and failure alert.

20. Reports & Exports

- Daily Consolidation Matching Summary
- Candidate / Excluded / Single Order Analysis
- Consolidation Batch and Split History
- Consolidated Pick Report A4
- Daily Order Progress
- Picker Productivity by Day / Shift / Zone Contribution
- Assignment Batch Performance / Override
- Warning / Overdue / Critical Detail
- Short Pick Reason Analysis
- Picking and Verification Backlog Aging
- Admin Verification Time
- Historical Weekly / Monthly Trend
- Excel export with filter criteria and generated timestamp

20.1 Weekly Productivity Export to Google Sheets

Purpose: move completed productivity history out of the transactional database every week while keeping a business-readable historical archive.

- Export grain: one row per Order productivity result; include Order No, Original Order Date, Store, Picker, Assigned Time, Picker Completed Time, Planned Pieces, Actual Pieces, Cycle Minutes, Pieces/Hour, Completion Result, Short Pick Reason, backlog indicators and relevant Zone contribution summary.

- Default schedule: weekly after operational week close. Day/time, timezone, Drive folder and file naming are configuration values; default timezone = Asia/Bangkok.
- Output method: Google Sheets API using a service account or organization-approved OAuth identity. Credentials are stored in secure environment/secret storage, never in source code or database plain text.
- Output file: create a new spreadsheet per week named BTM_Productivity_YYYY-Www to avoid Google Sheets cell-limit growth in one long-running workbook.
- Export control totals: row count, total planned pieces, total actual pieces, completed orders and short-pick orders must reconcile with database totals for the exported week.
- Export status: PENDING → RUNNING → SUCCESS / FAILED. Failed exports retry automatically and generate an alert for authorized users.

20.2 Transaction Data Retention / Week -1

- Default transaction_retention_days = 7. This value is configurable and not hard-coded.
- Purge applies to high-volume historical transaction tables such as imported order snapshots, order lines no longer active, matching candidates/history, completed assignment execution details and derived dashboard aggregates older than retention.
- Purge does not delete active/in-progress orders, users, roles, Location Master, Zone data, configuration, reason masters, or records required to operate the current system.
- Safety gate: data for a completed week cannot be purged until the corresponding weekly Google Sheets export status is SUCCESS and reconciliation passed.
- Purge job records purge date, covered period, row counts by table, export reference, executed-by/system job ID and result in housekeeping audit log.

21. Data Model & Key Relationships

Entity	Key Fields / Purpose
orders	order_id, order_no, original_date, store_code, planned_pieces, unique_sku, status
order_lines	line_id, order_id, sku, bin_code, qty, zone_code, pick_sequence
locations	bin_code, zone, aisle, side_pair, direction, bay, level, block, pick_sequence, active
consolidation_batches	consol_batch_id, order_date, priority, match_pct, stores, orders, sku, pieces, config_version, status
consolidation_orders	consol_batch_id, order_id, sequence
assignment_batches	assignment_batch_id, picker_id, admin_id, assigned_time, planned_pieces, workload_status, config_version
assignment_orders	assignment_batch_id, order_id, sequence, source_type/source_id
picker_completions	order_id, picker_completed_time, actual_pieces, result, short_reason, remark
admin_verifications	order_id, admin_id, decision, verified_time, reject_reason
employees/users	user_id, employee_id, name, role, scope, active
configuration	key, value, scope, effective_date, version, active
status_history	entity_type, entity_id, old/new status, user, timestamp, reason
audit_logs	user, action, entity, before/after, timestamp

Recommended indexes: order_no + original_date; order status; order_lines(bin_code); order_lines(sku); zone_code; consolidation status/date; picker_id; assigned_time; picker_completed_time; assignment_batch_id.

22. API / Integration Concept

Interface	Initial	Future
Order Import	Excel/CSV upload	WMS API / scheduled transfer
Location Master	Excel upload / DB master	Master API
Authentication	Local / enterprise login option	SSO / directory integration
Consolidation Output	Print/PDF + DB status	PDA task API
Picking Status	Web/PDA action	WMS event/API sync
Notifications	In-app	LINE / Email / Teams where approved
Weekly Productivity Archive	Google Sheets API / Google Drive folder	Scheduled weekly export with reconciliation and retry

Suggested internal API domains

- /orders, /imports, /locations, /matching, /consolidation-batches, /pick-reports, /assignment-batches, /picker-completions, /verifications, /dashboards, /configurations, /users, /audit

23. Security, Audit & Data Integrity

- RBAC at both menu and action level.
- Warehouse/Zone access scope where required.
- Override requires role + reason + audit.
- Batch generation/release must use transaction/locking to prevent the same order from being released twice.
- Configuration version must be recorded on created batches.
- Import source file metadata and row-level error reason retained.
- Admin verification decision, Short Pick and status changes are immutable history records even if current state changes.
- Sensitive credentials must not be stored in plain text; use secure hashing/SSO as appropriate.

24. Acceptance Criteria / UAT

ID	Scenario / Expected Result
UAT-01	Import the same WMS file twice; no duplicated order/line QTY.
UAT-02	One SKU appears in two Bin Codes in the same order; unique SKU count = 1 but report contains two physical pick lines.
UAT-03	Unknown Bin Code is blocked from released pick report and appears in error queue.
UAT-04	Exact same SKU signature creates P1 candidate without pairwise matching.
UAT-05	Order above configured Maximum Unique SKU is excluded to Single Order.
UAT-06	20-store perfect match splits to capacity-compliant batches and each meets min pieces.
UAT-07	A4 report sorts by composite Pick Sequence and barcodes are scannable.
UAT-08	Assignment pieces accumulate instantly and color follows configuration.
UAT-09	Timer starts only at Confirm Assignment.
UAT-10	Picker closes order; KPI stops at Picker Completed.
UAT-11	Short Pick requires actual/short/reason and remark where configured.
UAT-12	Admin confirmation makes Final Closed; reject returns to correction.

ID	Scenario / Expected Result
UAT-13	Order with Shelf + Rack lines appears as one order productivity record and contributes pieces to both Zones.
UAT-14	Control Tower Total Orders remains unique and is not equal to sum of Orders Touching Zone.
UAT-15	Picking and Verification Backlog are separated correctly across day end.
UAT-16	Role permission hides/blocks unauthorized menu/action.
UAT-17	Configuration change is effective-dated and does not change in-progress batch rules.
UAT-18	Historical search and Excel export works for weekly Google Sheets archive.

25. Implementation Boundary

Implementation Responsibility	Requirement Boundary	Required Outcome
Product Requirement	Business rules, shared order core, module behavior, UX/UI standard, integration rules, data retention and UAT criteria	Requirements in this document remain the source of truth
Programmer / AI Coding	Technical decomposition, task sequence, code architecture within stated constraints, effort estimation and release packaging	Implementation plan does not change business logic
Quality / UAT	Functional test, performance test, security test, print/barcode validation and reconciliation testing	All Must requirements and acceptance criteria pass
Deployment / Operations	Environment setup, secrets, scheduled jobs, backup, monitoring and production support	Stable production operation and recoverability
Future Enhancement	PDA consolidation tasks, WMS API automation, notifications, forecasting and advanced intelligence	Added only without breaking approved Product behavior

This document defines Product behavior, data rules, UX/UI standards, integrations, retention and acceptance criteria only.

Task decomposition, coding sequence, effort estimation and release planning are implementation responsibilities of the Programmer / AI coding workflow and must not change the business rules defined in this document.

- Implementation may be delivered incrementally as long as Shared Order Core remains the single source of truth and module behavior remains backward compatible with the approved Product Requirement.

26. Defined Business Rules & Technical Defaults

Item	Defined Rule / Technical Default
Operational day / Order date	Order Consolidation defaults to D-1. Backlog and productivity use the operational date rules in this requirement.
Assignment capacity	Pieces target/tolerance and optional maximum order count are configuration-driven.
Barcode symbology	Code128 is the default for Bin and SKU barcodes unless an existing WMS barcode standard is supplied.
Zone derivation	Zone is derived from WMS Bin Code through Location Master. Division is not used.
Multiple orders per store	Allowed. Distinct Store and Order counts are tracked separately.
Location stock choice	Not performed by this Product. WMS-provided Bin Code is the authoritative physical pick location.
Productivity grain	Productivity is measured at Order level; Zone contribution is derived from Order Lines / Bin Code.
Weekly productivity archive	A new Google Spreadsheet is generated weekly in a configured Drive folder; export must reconcile row count and totals before status = SUCCESS.
Database retention	High-volume transaction data older than 7 days is purged only after the related weekly export is SUCCESS. Master/config/user records are excluded from purge.
Implementation sequencing	Programmer / AI coding team determines technical decomposition, coding sequence and release packaging; this is outside the Product Requirement.

All values below are implementation defaults or configuration rules. The document contains no pending business question and does not require confirmation before development starts.

Appendix A. Field Dictionary - Core

Field	Type Concept	Notes
order_no	string	WMS order identifier
original_order_date	date	Matching partition / historical key
store_code	string	Store destination
sku	string	Matching unique SKU basis
bin_code	string	Actual WMS pick location
qty	decimal/int	Planned pieces
zone_code	string derived	From Location Master by Bin
pick_sequence	string/numeric composite	Physical sort key
consolidation_batch_id	string	Optimization batch
assignment_batch_id	string	Productivity work package
picker_completed_time	datetime	Stops order KPI
actual_picked_pieces	numeric	Completion KPI
short_reason	master code	Required for short pick

Appendix B. Status Matrix

Entity	Representative Statuses
Order	New, Assigned, In Progress, Warning, Overdue, Critical, Picker Completed, Waiting Admin, Final Closed, Backlog, Cancelled
Consolidation Batch	Candidate, Review, Approved, Report Released, Picking, At Consolidation, Sorting, Completed, Cancelled
Assignment Batch	Draft, Assigned, In Progress, Completed, Over Capacity Override, Cancelled
Import	Uploaded, Validating, Completed, Completed with Errors, Failed

Appendix C. Source / Baseline Notes

This requirement consolidates the existing Picking Productivity & Operation Control Tower baseline with the latest Order Consolidation rules: Shared Order Core, no Division-to-Zone mapping, Zone derived from WMS Bin Code through Location Master, order-level productivity, multi-Zone order contribution, WMS Bin Code as the authoritative pick location, and physical Pick Sequence ordering.

End of document